



UNIVERSITY OF THE PUNJAB

B.S. in Computer Science Third Semester : Spring 2024

Subject: Calculus and Analytical Geometry

Paper: MS-251 N

Roll No. [redacted]
Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Define the following:

(6x5=30)

- I. If $f(x) = x + 5$, $g(x) = x^2 - 3$. Find formula for $f(g(x))$. Also calculate $f(g(0))$?
- ✓II. 3 Evaluate $\lim_{x \rightarrow 1} \frac{x^2 + x - 2}{x^2 - x}$?
- III. Find an equation for the normal line to the curve at the given point. $x^2 - xy + y^2 = 7$, $(-1, 2)$.
- IV. Find the intervals of concavity and inflection points of $f(x) = x + \sin x$ for $0 \leq x \leq 2\pi$.
- ✓V. Evaluate the integral $\int \cos^2 x \, dx$?
- ✓VI. Find parametric equations for line through $P(-3, 2, -3)$ and $Q(1, -1, 4)$?

Q.2. Answer the following questions.

(3x10=30)

- I. Find an equation for the tangent line to the curve at the given point. Then sketch the curve and tangent together. $f(x) = x^3$, $(-2, -8)$.
 $\int \cos^3 x = \cos x \sin x + \int \sin^2 x = \cos x \sin x - \frac{1}{2} \cos 2x + \frac{1}{2} \cos x + C$
- II. a) Find two positive numbers whose sum is 20 and whose product is as large as possible.
b) Find all critical points of $f(x) = x^{4/3} - 4x^{1/3}$.
- III. Find the point of intersection of the lines $L1: x = 1 + 2t, y = 2 + 3t, z = 3 + 4t$
 $L2: x = 2 + t, y = 4 + 2t, z = -1 - 4t$ and then find the plane determined by the lines?